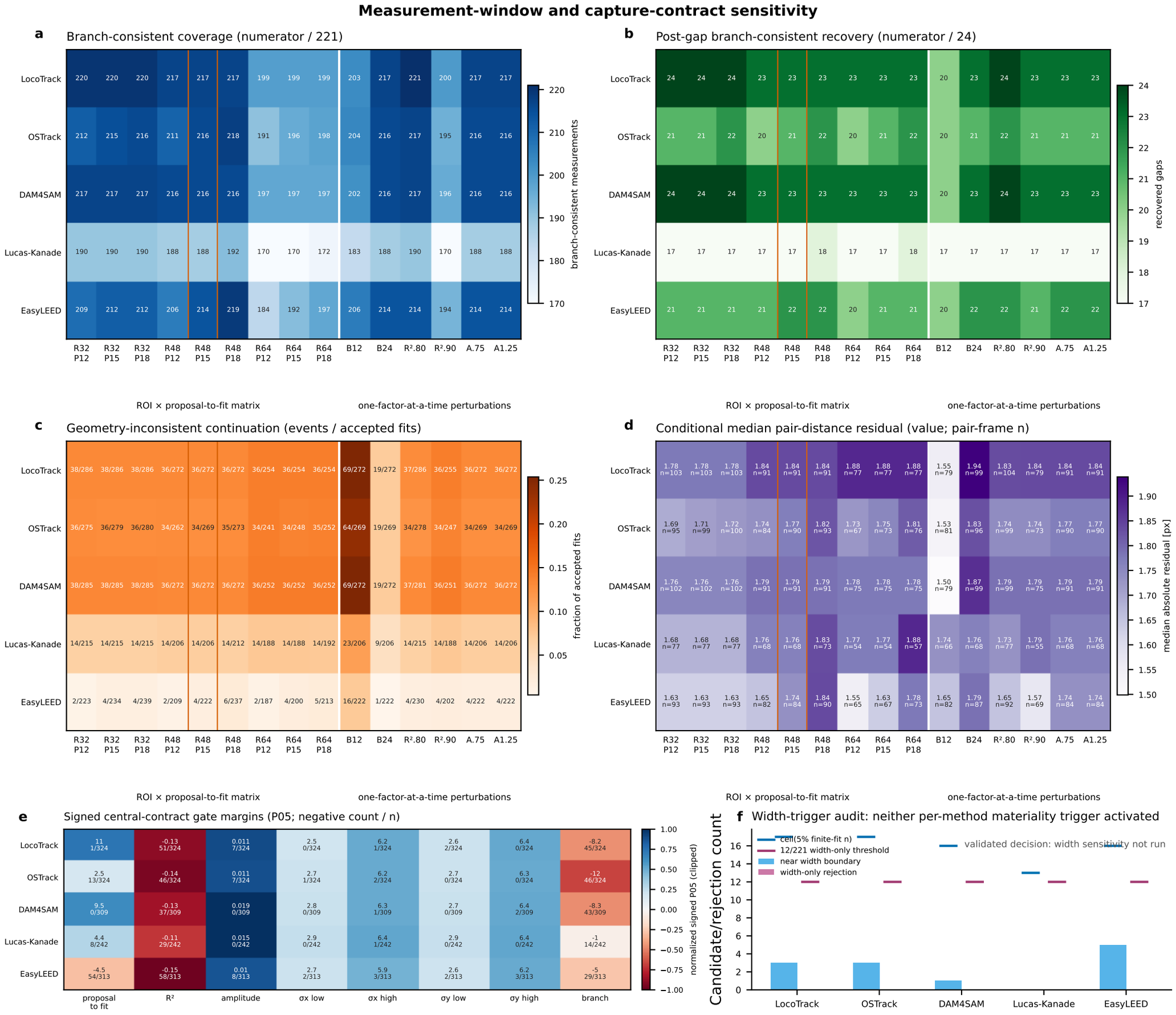
Measurement-window and capture-contract sensitivity
a
Branch-consistent coverage (numerator / 221)
LocoTrack 220 220 220 217 217 217 199 199 199 203 217 221 200 217 217
OSTrack 212 215 216 211 216 218 191 196 198 204 216 217 195 216 216
DAM4SAM 217 217 217 216 216 216 197 197 197 202 216 217 196 216 216
Lucas-Kanade 190 190 190 188 188 192 170 170 172 183 188 190 170 188 188
EasyLEED 209 212 212 206 214 219 184 192 197 206 214 214 194 214 214
branch-consistent measurements
220
210
200
190
180
170
R32 P12
R32 P15
R32 P18
R48 P12
R48 P15
R48 P18
R64 P12
R64 P15
R64 P18
B12
B24
R².80
R².90
A.75
A1.25
ROI × proposal-to-fit matrix
one-factor-at-a-time perturbations
b
Post-gap branch-consistent recovery (numerator / 24)
LocoTrack 24 24 24 23 23 23 23 23 23 20 23 24 23 23 23
OSTrack 21 21 22 20 21 22 20 21 22 20 21 22 21 21 21
DAM4SAM 24 24 24 23 23 23 23 23 23 20 23 24 23 23 23
Lucas-Kanade 17 17 17 17 17 18 17 17 18 17 17 17 17 17 17
EasyLEED 21 21 21 21 22 22 20 21 21 20 22 22 21 22 22
recovered gaps
24
23
22
21
20
19
18
17
ROI × proposal-to-fit matrix
one-factor-at-a-time perturbations
c
Geometry-inconsistent continuation (events / accepted fits)
LocoTrack 38/286 38/286 38/286 36/272 36/272 36/272 36/254 36/254 36/254 69/272 19/272 37/286 36/255 36/272 36/272
OSTrack 36/275 36/279 36/280 34/262 34/269 35/273 34/241 34/248 35/252 64/269 19/269 34/278 34/247 34/269 34/269
DAM4SAM 38/285 38/285 38/285 36/272 36/272 36/272 36/252 36/252 36/252 69/272 19/272 37/281 36/251 36/272 36/272
Lucas-Kanade 14/215 14/215 14/215 14/206 14/206 14/212 14/188 14/188 14/192 23/206 9/206 14/215 14/188 14/206 14/206
EasyLEED 2/223 4/234 4/239 2/209 4/222 6/237 2/187 4/200 5/213 16/222 1/222 4/230 4/202 4/222 4/222
fraction of accepted fits
0.25
0.20
0.15
0.10
0.05
R32 P12
R32 P15
R32 P18
R48 P12
R48 P15
R48 P18
R64 P12
R64 P15
R64 P18
B12
B24
R².80
R².90
A.75
A1.25
ROI × proposal-to-fit matrix
one-factor-at-a-time perturbations
d
Conditional median pair-distance residual (value; pair-frame n)
LocoTrack 1.78 n=103 1.78 n=103 1.78 n=103 1.84 n=91 1.84 n=91 1.84 n=91 1.88 n=77 1.88 n=77 1.88 n=77 1.55 n=79 1.94 n=99 1.83 n=104 1.84 n=79 1.84 n=91 1.84 n=91
OSTrack 1.69 n=95 1.71 n=99 1.72 n=100 1.74 n=84 1.77 n=90 1.82 n=93 1.73 n=67 1.75 n=73 1.81 n=76 1.53 n=81 1.83 n=96 1.74 n=99 1.74 n=73 1.77 n=90 1.77 n=90
DAM4SAM 1.76 n=102 1.76 n=102 1.76 n=102 1.79 n=91 1.79 n=91 1.79 n=91 1.78 n=75 1.78 n=75 1.78 n=75 1.50 n=79 1.87 n=99 1.79 n=99 1.79 n=75 1.79 n=91 1.79 n=91
Lucas-Kanade 1.68 n=77 1.68 n=77 1.68 n=77 1.76 n=68 1.76 n=68 1.83 n=73 1.77 n=54 1.77 n=54 1.88 n=57 1.74 n=66 1.76 n=68 1.73 n=77 1.79 n=55 1.76 n=68 1.76 n=68
EasyLEED 1.63 n=93 1.63 n=93 1.63 n=93 1.65 n=82 1.74 n=84 1.84 n=90 1.55 n=65 1.63 n=67 1.78 n=73 1.65 n=82 1.79 n=87 1.65 n=92 1.57 n=69 1.74 n=84 1.74 n=84
median absolute residual [px]
1.90
1.85
1.80
1.75
1.70
1.65
1.60
1.55
1.50
R32 P12
R32 P15
R32 P18
R48 P12
R48 P15
R48 P18
R64 P12
R64 P15
R64 P18
B12
B24
R².80
R².90
A.75
A1.25
ROI × proposal-to-fit matrix
one-factor-at-a-time perturbations
e
Signed central-contract gate margins (P05; negative count / n)
LocoTrack 11 1/324 -0.13 51/324 0.011 7/324 2.5 0/324 6.2 1/324 2.6 2/324 6.4 0/324 -8.2 45/324
OSTrack 2.5 13/324 -0.14 46/324 0.011 7/324 2.7 1/324 6.2 2/324 2.7 0/324 6.3 0/324 -12 46/324
DAM4SAM 9.5 0/309 -0.13 37/309 0.019 0/309 2.8 0/309 6.3 1/309 2.7 0/309 6.4 2/309 -8.3 43/309
Lucas-Kanade 4.4 8/242 -0.11 29/242 0.015 0/242 2.9 0/242 6.4 1/242 2.9 0/242 6.4 0/242 -1 14/242
EasyLEED -4.5 54/313 -0.15 58/313 0.01 8/313 2.7 2/313 5.9 3/313 2.6 2/313 6.2 3/313 -5 29/313
proposal to fit
R²
amplitude
σx low
σx high
σy low
σy high
branch
normalized signed P05 (clipped)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
ROI × proposal-to-fit matrix
one-factor-at-a-time perturbations
f
Width-trigger audit: neither per-method materiality trigger activated
cell(5% finite-fit n)
12/221 width-only threshold
near width boundary
width-only rejection
validated decision: width sensitivity not run
Candidate/rejection count
16
14
12
10
8
6
4
2
0
LocoTrack
OSTrack
DAM4SAM
Lucas-Kanade
EasyLEED